



Birla Institute of Technology & Science, Pilani

Work-Integrated Learning Programmes Division

MTech in Artificial Intelligence and Machine Learning

S2_2025-2026, AIMLCZG557/AECLZG557

Assignment 1 – PS10 - Water Quality - [Weightage 13%]

Read through this entire document very carefully before you start!

Problem Statement

Water quality data is provided along with this file and contains the following attributes
Attributes Information:

1. ph: pH of 1. water (0 to 14).
2. Hardness: Capacity of water to precipitate soap in mg/L.
3. Solids: Total dissolved solids in ppm.
4. Chloramines: Amount of Chloramines in ppm.
5. Sulfate: Amount of Sulfates dissolved in mg/L.
6. Conductivity: Electrical conductivity of water in $\mu\text{S}/\text{cm}$.
7. Organic_carbon: Amount of organic carbon in ppm.
8. Trihalomethanes: Amount of Trihalomethanes in $\mu\text{g}/\text{L}$.
9. Turbidity: Measure of light emitting property of water in NTU.
10. Potability: Indicates if water is safe for human consumption. Potable -1 and Not potable -0

You are required to do the following:

Question 1: Python

1. Construct a Bayesian Belief Network for the given data.

Use appropriate methods to predict the following:

2. Predict the water quality for the following data:

ph	Hardness	Solids	Chloramines	Sulfate	Conductivity	Turbidity
3.72	204.89	20791.32	7.3	368.5	564.30	2.96

3. Infer the probability for the data with the following properties:

Hardness	Solids	Chloramines	Sulfate	Conductivity	Organic carbon	Trihalomethanes	Turbidity	Potability
248.	28749	7.5	393	283	13.78	84.6	2.67	1

4. Find the probability of the quality of water being good and the attributes take the following values: low ph, high in hardness, with high presence of solids, and other chemicals.

Sample Input

1. Predict the water quality for the following data:

ph	Hardness	Solids	Chloramines	Sulfate	Conductivity	Turbidity
3.72	204.89	20791.32	7.3	368.5	564.30	2.96

Note that the input/output data shown here is only for understanding and testing, the actual file used for evaluation will be different.

Sample Output

Potability	0	1
Probability	0.571838	0.428162

Probability of the water being good is 42.81% when considered low ph value and remaining variables high value from given dataset

Note that the input/output data shown here is only for understanding and testing, the actual file used for evaluation will be different.

Display the output in outputPSXX.txt.

Sample Input 2

1. Infer the probability for the data with the following properties:

Hardness	Solids	Chloramines	Sulfate	Conductivity	Organic carbon	Trihalomethanes	Turbidity	Potability
248.	28749	7.5	393	283	13.78	84.6	2.67	1

Note that the input/output data shown here is only for understanding and testing, the actual file used for evaluation will be different.

Sample Output 2

Potability	0	1
Probability	0.65359	0.34751

Probability of the water being good is 34.75% when the given attribute values are given from the dataset

Note that the input/output data shown here is only for understanding and testing, the actual file used for evaluation will be different.

Deliverables:

- PDF document **designPSXX_<group id>.pdf** detailing your solution.
- **[Group id] _Contribution.xlsx** mentioning the contribution of each student in terms of percentage of work done. Columns must be “Student Registration Number”, “Name”, “Percentage of contribution out of 100%”. If a student did not contribute at all, it will be 0%, if all contributed then 100% for all.
- **inputPSXX.txt** file used for testing
- **outputPSXX.txt** file generated while testing
- .py file containing the python code. Do not fragment your code into multiple files
- Zip all of the above files including the design document in a folder with the name:
 - **[Group id] _A2_PSXX.zip** and submit the zipped file.
 - Group Id should be given as **Gxxx** where **xxx** is your group number. For example, if your group is 26, then you will enter **G026** as your group id.

Instructions:

- It is compulsory to make use of the data structure(s) / algorithms mentioned in the problem statement.
- Ensure that all data structure insert and delete operations throw appropriate messages when their capacity is empty or full. Also ensure basic error handling is implemented.
- For the purposes of testing, you may implement some functions to print the data structures or other test data. But all such functions must be commented before submission.
- Make sure that you read, understand, and follow all the instructions
- Ensure that the input, prompt and output file guidelines are adhered to. Deviations from the mentioned formats will not be entertained.
- The input, prompt and output samples shown here are only a representation of the syntax to be used. Actual files used to evaluate the submissions will be different. Hence, do not hard code any values into the code.
- Please note that the design document must include:
 - One alternate way of modeling the problem with the performance implications.
 - Writing a good technical report and well documented code is an art. Your report cannot exceed 4 pages. Your code must be modular and quite well documented.
 - You may ask queries [in this dedicated discussion section](#). Beware that only hints will

be provided and queries asked **in other channels like MS Teams, emails, Taxila inbox will not be responded to.**

Deadline

1. The strict deadline for submission of the assignment is **17th Aug 2026 11:55 PM IST (Monday).**
2. The deadline has been set considering extra days from the regular duration in order to accommodate any challenges you might face. No further extensions will be entertained.
3. Late submissions will be evaluated as below. Submissions made between:
 - a. 17th Aug 2026 11:56PM to 18th Aug 2026 11:55PM → Deduct 2M for late submission and evaluation for 10 marks.
 - b. 18th Aug 2026 11:56PM to 19th June 2026 11:55PM → Deduct 6M for late submission and evaluation for 6marks.
 - c. Beyond 19th Aug 2026 11:55PM → No evaluations, 0 marks will be awarded.

How to submit

- This is a group assignment.
- Each group has to make one submission (only one, no resubmission) of solutions.
- Each group should zip all the deliverables in one zip file and name the zipped file as [Group id] _A2_PSXX.zip and submit the zipped file.
- Group Id should be given as Gxxx where xxx is your group number. For example, if your group is 26, then you will enter G026 as your group id.
- Assignments should be submitted via Taxila > Assignment section. Assignments submitted via other means like email etc. will not be graded.

Evaluation

- The assignment carries **13 Marks.**
- Grading will depend on:
- Fully executable code with all functionality working as expected
 - Well-structured and commented code
 - Accuracy of the design document.
 - Every bug in the functionality will have negative marking.
 - Marks will be deducted if your program fails to read the input file used for evaluation due to change / deviation from the required syntax.
- We encourage students to take the upcoming assignments and examinations seriously and submit only original work. Please note that plagiarism in assignments will be taken seriously. Refer to the detailed policy [here](#).
- Source code files which contain compilation errors will get at most 25% of the value of that question.

Readings

Artificial Intelligence: A Modern Approach Textbook by Peter Norvig and Stuart J. Russell, 4th Edition.